Dukyong Kim

[Google Scholar] [LinkedIn] [ORCID]

Education

Ph.D. **Northwestern University (Evanston Campus)** | Sep. 2025 – current
Department of Mechanical Engineering — Illinois, U.S.
Fast-AM Laboratory (PI: Prof. Tao Sun)

- Cumulative GPA: 3.92/4.0
- Predictive Science and Engineering Design (PS&ED) Cluster Fellow (2026–2027) - Selected for the interdisciplinary Northwestern University Graduate School fellowship.

M.S. **Hanyang University (Seoul Campus)** | Mar. 2023 – Feb. 2025
Department of Mechanical Convergence Engineering — Seoul, South Korea
Smart Manufacturing Laboratory (PI: Prof. Seung Hwan Lee)

- Cumulative GPA: 4.0/4.0

B.S. **Hanyang University (Seoul Campus)** | Mar. 2017 – Feb. 2023
Department of Mechanical Engineering — Seoul, South Korea

- Cumulative GPA: 3.84/4.0 (*Summa Cum Laude*)
  Major GPA: 3.84/4.0, 3rd-4th year GPA: 3.92/4.0
- Supported by full National Science and Engineering Undergraduate Scholarship by the Korean Government

Employment

Advanced Photon Source, Argonne National Laboratory | Aug. 2026 – current
XSD-IMG group — Illinois, U.S.

- Visiting Graduate Student

Hanyang University | Mar. 2025 – Jun. 2025
The Research Institute of Industrial Science — Seoul, South Korea

- Graduate Researcher (4 month)

Korea Insurance Development Institute | Jun. 2025 – July. 2025
Korea Automobile Insurance Repair Research & Training Center — Seoul, South Korea

- Automobile Accident Classification Analysis

Research Interests

Research keywords

Metal additive manufacturing	Synchrotron x-ray imaging	Machine learning	Software development	Laser spallation test

Operando synchrotron x-ray imaging of laser direct energy deposition (L-DED) process

- Collaborated with Advanced Photon Source, Argonne National Lab (Lemont, US) to conduct operando synchrotron x-ray high-speed x-ray imaging of multi-material stranded wire laser direct energy deposition process.

Laser spallation test to quantify interfacial adhesion strength thin film

- Pulsed laser-induced shock wave test for quantifying interfacial adhesion properties of thin films

Multi-axial powder laser direct energy deposition process

- Metal-based additive manufacturing for fabricating intricate structures under laser nozzle tilting

Machine learning for prediction and object tracking

- Developed full-field x-ray image analysis software that is open source, which incorporates up-to-date machine learning models for object annotation and tracking [GitHub link]
- Physics-informed Gaussian process regression model for fatigue life prediction of welded joints
- Physics-informed neural network for lifespan prediction of proton-exchange membrane fuel cells

Publications

International Journal

1. **First author** | Dukyong Kim*, Tao Sun* (2026/07) *corresponding author

   "SMAXI: A machine learning-powered open-source software for multidimensional full-field x-ray image analysis", *Journal of Synchrotron Radiation* (2026/09) [link]

2. **Co-author** | Junmyoung Jang, Taehwan Ko, Jaeheon Lee, Dukyong Kim, Hwani Hwang, Taesoon Jeong, Heuisu Kim, Seung Hwan Lee* (2025/07)

   "Energy-Assisted additive manufacturing for weldable and non-weldable Ni-based superalloys: a review", *Virtual and Physical Prototyping* [link]

3. **First author |** Young-Min Ju†, Dukyong Kim†, Se-Min Lee, Daewoong Lee, Yeontaek Hwang, Seung Hwan Lee*, and Hak-Sung Kim* (2025/03)

   "Measurement of interfacial adhesion strength of Copper- Silicon based dielectric interfaces via laser spallation test", *Applied Surface Science Advances*, (†: equally contributed) [link]

4. **First author |** Taehwan Ko†, Dukyong Kim†, Jaewoong Park, and Seung Hwan Lee* (2025/03)

   "Physics-informed neural network for long-term prognostics of proton exchange membrane fuel cell", *Applied Energy* (†: equally contributed) [link]

5. **First author |** Dukyong Kim†, Dong-Yoon Kim†, Taehwan Ko†, and Seung Hwan Lee* (2024/10)

   "Physics-informed Gaussian process regression model for predicting the fatigue life of welded joints", *International Journal of Fatigue* (†: equally contributed) [link]

6. **First author |** Dukyong Kim, Taehwan Ko, and Seung Hwan Lee* (2023/05)

   "Study on the influence of processing variables under various tilting angles of laser nozzle and substrate on the deposition quality of the multi-axis laser metal deposition", *Journal of Laser Applications* [link]

Conference paper

1. **Co-author |** "Review on welding process monitoring based on deep learning using time-series data",

Journal of Welding and Joining, (2024/07) [link]

2. **Co-author |** "Review of recent additive manufacturing and welding research with application of physics-informed neural networks", *Journal of Welding and Joining*, (2024/07) [link]
3. **Co-author |** "Measurement of the interfacial strength of thin metal film by laser spallation method for advanced wafer level package", *2024 IEEE 74th Electronic Components and Technology Conference* (2024/05) [link]

Conference

1. **Presenting author (Poster) |** "Multi-material stranded-wire laser directed energy deposition for in-space manufacturing", *The Minerals, Metals, and Materials (TMS) 156th Annual Meeting,* 05.14. 2027
2. **Presenting author (Oral) |** "Machine learning accelerated open-source software for analyzing operando full-field x-ray image of additive manufacturing process", *AI-Empowered Simulations & CAE Applications via Engineering Software 2.0 and 3.0 2026,* 09.11. 2026
3. **Presenting author (Oral) |** "Machine learning-powered open-source full-field x-ray image analysis software", *Advanced Photon Source (APS) User Science Seminar*, 03.27. 2026
4. **Presenting author (Oral) |** "Physics-informed Gaussian process regression model for predicting the fatigue life of welded joints", *The Korean Welding and Joining Society*, 10.17. 2024
5. **Presenting author (Oral) |** "Laser spallation test for measuring interface strength of thin films", *21st International Symposium on Microelectronics and Packaging* (ISMP), 10.25. 2023
6. **Presenting author (Oral) |** "Measurement of the interfacial adhesion of thin films by laser spallation test", *The Korean Welding and Joining Society*, 10.12. 2023
7. **Co-author |** "Operando X-ray Imaging of Porosity Formation During Wire-Laser Directed Energy Deposition", *The Minerals, Metals, and Materials (TMS) 156th Annual Meeting,* 05.14. 2027
8. **Co-author |** "Domain adaptation from bare-late to powder-bed data for real-time keyhole porosity detection in laser powder bed fusion", *ASTM International Conference on Advanced Manufacturing (ICAM 2026),* 10.01. 2026
9. **Co-author | "**Direct observation of acoustic cavitation in laser additive manufacturing of Aluminum using high-speed synchrotron X-ray imaging", *Acoustical Society of America (ASA) conference*, 05.11. 2026
10. **Co- author |** "From bubble to pore: Acoustic signatures of keyhole cavity collapse", *Acoustical Society of America (ASA) conference*, 05.11. 2026
11. **Co-author | "**X-Ray-Informed Acoustic Emission Modeling of Melt Pool and Keyhole Dynamics in Laser Powder Bed Fusion", *Acoustical Society of America (ASA) conference*, 05.11. 2026
12. **Co-author | "**Evaluation of Interfacial Adhesion Strength Between Copper and Silicon Nitride Thin Films Deposited by LPCVD and PECVD Using Laser Spallation Test", *Korean Society for Precision Engineering*, 05.14. 2025
13. **Co-author | "**Laser spallation test for quantifying the interfacial adhesion between copper and silicon nitride thin films deposited by low-pressure chemical vapor deposition and plasma-enhanced chemical vapor deposition process", *The 10th International Conference on Manufacturing, Machine Design and Tribology*, 04.24. 2025
14. **Co-author |** "Laser Spallation Test for Measurement of the Interfacial Adhesion of LPCVD and PECVD Silicon Nitride/Copper Films", *22nd International Symposium on Microelectronics and Packaging* (ISMP), 11.07. 2024
15. **Co-author |** "Investigation of adhesion at LPCVD and PECVD silicon nitride/copper film interface by laser spallation technique", *The Korean Society of Mechanical Engineers*, 06.27. 2024
16. **Co-author |** "Measurement of the interfacial properties of thin metal film by laser spallation method for advanced wafer level package", IEEE *74th Electronic Components and Technology Conference*, 05.28. 2024
17. **Co-author |** "Laser spallation test for measurement of interfacial properties of thin film", *The Korean Microelectronic and Packaging Society*, 04.03. 2024

Research Experience

Graduate student | Northwestern University
(experiment conducted with APS, Argonne National lab) | Sep. 2025 – current

- Abnormal Columnar-to-Equiaxed Transition in Metals Processed Under Non-Equilibrium Conditions in Laser Additive Manufacturing (https://doi.org/10.46936/APS-195894/60018199) (09.22.26)
- Establishing a Data-Driven Predictive Framework for Spatter-Induced Porosity in Laser Additive Manufacturing through High-Throughput Synchrotron Micro-CT (https://doi.org/10.46936/APS-195705/60018025) (09.11.26)
- Micro computed tomography characterization of spatter-induced porosity in additive manufacturing Ti-6Al-4V (https://doi.org/10.46936/APS-194891/60017462) (07.10.26)
- Operando X-ray Imaging of Laser Directed Energy Deposition Process with Stranded Multi-Material Wires (https://doi.org/10.46936/APS-193419/60016438) (03.30.26)
- Abnormal Columnar-to-Equiaxed Transition in Metals Processed Under Non-Equilibrium Conditions in Laser Additive Manufacturing (https://doi.org/10.46936/APS-193316/60016390) (03.25.26)
- Development of hybrid manufacturing process for building large-scale stress-free metallic parts (https://doi.org/10.46936/APS-191914/60015668) (12.19.25)
- Simultaneous x-ray imaging and ultrasound detection of keyhole porosity in laser powder bed fusion (https://doi.org/10.46936/APS-191010/60014768) (10.20.25)

Graduate student | Hanyang University | Mar. 2023 – Feb. 2025

- Contributed toward building the nation's first laser spallation system
  → *Master's thesis*: Measurement of the interfacial adhesion strength of low-pressure chemical vapor deposition processed Silicon Nitride and Copper thin films by laser spallation test (UCI: I804:11062-200000826436)
- Developed physics-informed machine learning models for prognostics and health management (PHM)

Undergraduate research student | Hanyang University | Feb. 2022 – Mar.2023

- Explored the effect of laser nozzle tilting and substrate inclination angle on the deposition quality of Inconel 625 fabricated by the multi-axis laser metal deposition (LMD) process.
  → *Undergraduate thesis*: "Development of geometry prediction model based on Gaussian process regression for evaluation of Inconel 625 formed by LMD method"

Research Projects

- Development of smart DPO (Design, Production, Operation) open platform for fuel cell system, 2021-2025 (48 months), researcher, *Ministry of Trade, Industry and Energy*, Korea
- Study on the technology of fault diagnosis and predictive maintenance of the fuel cell system using PHM (Prognostics and Health Management), 2021-2023 (24 months), *Hanyang University Industry-University Cooperation Foundation*, Korea
- Analysis of the relationship between the laser nozzle angle and the deposition quality, 2022 (6 months), *DN Solutions Incorporate*, Korea
- Development of prediction technology of packaging process and reliability based on Die MLM thin film interface property and laser behavior, 2022-2024 (24 months), *SK Hynix Incorporation*, Korea

Teaching experience

- Teaching assistant: Characterization of Materials (2026.09)
- ELP (English Language Programs) Foundations Online Peer Mentor (2026.06-7)
- Teaching assistant: Linear algebra and its applications (2023.03, 2024.03)

Fellowship

- Predictive Science and Engineering Design (PS&ED) Cluster Fellow | Northwestern University (2026–2027)
  Awarded a competitive, interdisciplinary fellowship focused on integrating data science, physics-based simulation, and uncertainty quantification into advanced engineering design methodologies.

Scholarship

- Scholarship of BK21 FOUR (Brain Korea 21) Research Assistant | Sep. 2023 – Dec. 2023
- HY-IN Scholarship, Hanyang University (master's program scholarship) | Mar. 2023 – Dec. 2024
- Scholarship of Hanyang College of Engineering International Student Ambassador | Mar. 2021 – Feb. 2022
- National Science and Engineering Undergraduate Scholarship | Mar. 2017 – Feb. 2023

Awards and extracurricular activities

Awards

- Best Oral Presentation Award, *Conference of The Korean Welding and Joining Society* (2024/10)
- Undergraduate Academic Excellence award, *Summa Cum Laude* (2023/02)
- Hanyang English Essay Contest, Excellent Award (2020/12)

Extracurricular activities

- Social chair of Mechanical Engineering Graduate Student Society, Northwestern University (2026)
- Co-translation of "Manufacturing Engineering and Technology" by S. Kalpakjian (2023, 6 months)
- International student ambassador of Hanyang University College of Engineering (2021, 12 months)
- Club president of Hanyang University Material Science Dept. band club (2020, 12 months)

Professional skills

Language skills (Korean: Native, English: Native, Japanese: Intermediate)

- TOEFL – Feb. 2024

  Total score: 111/120, reading: 30/30, listening: 28/30, speaking: 25/30, writing: 28/30

- GRE general test – Feb. 2022

  Verbal reasoning: 160/170 (84%), quantitative reasoning: 170/170 (92%), analytical writing: 4.0/6.0 (59%)

Testing skills

- Metal additive manufacturing processes

  (laser powder bed fusion, powder-blown and wire-fed L-DED process)

- Laser optics (based on laser spallation test)

  (Q-switched Nd:YAG laser, disk laser, diode pumped solid state laser, Michelson interferometer)

- Universal testing machine (axial fatigue test, tensile shear test)

- Material characterization (SEM, EDS, EBSD analysis)

Software skills

- Python, MATLAB, NX (Siemens), Delmia (Dassault Systèmes), Adobe creative suite
- Abaqus, Flow3D, ANSYS
- Computer Specialist in Spreadsheet and Database – Level 1

Military service

- Republic of Korea Army (2018.03.12 - 2019.11.10, 20 months)